

Memorandum

To : Attached Distribution List

From : Yat-Sang Hung, Human Factors Engineering, USWC

Re: Preliminary Usability Results of Capacity Planner Tool Set
Release 1.0

Date: May 5, 1995

Attached please find the preliminary report for the usability studies of Capacity Planner Tool Set Release 1.0. We would like to take this chance to express our gratitude to the participants of the studies, the Development Team of Capacity Planner Tool Set, the System Operations Group and the Process Prototype Team for their support and assistance.

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**Capacity Provisioning
Capacity Planner Tool Set Release 1.0
Usability Studies
Preliminary Report**

**Human Factors Engineering
U S WEST Communications**

1.0 Introduction

The Capacity Planner Tool Set is an integrated knowledge-based software application supporting the Future Method of Operation (FMO) for Tactical Planning Process in Capacity Provisioning. It consists of the Capacity Analyzer (CA), the Architectures/Models/Configurations (AMC) Qualifier and the Plan Outliner (PO). In Release 1.0 of the Tool Set, only the Capacity Analyzer will be deployed. Subsequent releases will include the other two modules as well as enhancements to Capacity Analyzer.

The goal of this usability evaluation was to obtain feedback from Capacity Provisioning Tactical Planners and other users on the Capacity Planner Tool Set Release 1.0. This evaluation focused on the usability of Capacity Planner Tool Set as well as whether the information provided by the Capacity Analyzer of the Tool Set supports analysis of loop information.

The Capacity Planner Tool Set uses a graphical user interface designed to support tactical planning activities of Capacity Provisioning. Release 1.0 of the Tool Set enables the tactical planners to create new or open existing planning request for selected wire centers, view the overall hierarchical organization of the wire center selected (route, CSA and DA) and compare integrated forecasts of functional properties (FP) groups by location area to existing and planned network functional capabilities. It also allows the users to modify forecasts and capacity data as needed.

2.0 Method

Evaluations of the usability of the system were conducted in the Deer Creek Capacity Provisioning Facilities in conjunction with the User Acceptance Testing of the System Operations Group of Capacity Planner Tool Set. The participants were given a brief overview of the Tool Set before they conducted user acceptance testing given by System Operations. After the user acceptance testing, the participants completed a Usability Study Questionnaire, a User Profile Inventory as well as a short Questionnaire designed to assess their training needs. A total of 11 users (8 tactical planners, 2 prototype process team members and 1 design engineer) participated in the study.

3.0 Results

Some of the important results from the usability studies will be considered. Specific comments and suggestions from the usability study questionnaire provided valuable information and feedback so that continuous improvement of the system can be implemented. Specific recommendations for Capacity Planner Tool Set Release 1.0 will be included in the final report. Preliminary usability results are discussed below for the Capacity Planner Tool Set Release 1.0.

3.1 Limitations of Results

The results in this report should be interpreted with *caution* because of the adverse conditions under which the testing was conducted. The performance of the Capacity Planner Tool Set was less than optimal in the testing due to network and database problems. No user acceptance testings were conducted between April 27 and 28, 1995 because of hardware problems of the database server. During that period, the participants were shown the Capacity Planner Tool Set from the development machine. After the demonstration, they completed usability questionnaires, user profile inventory and the training needs questionnaires instead of going through the user acceptance testing phase.

Since the participants have not received any formal Capacity Planner Tool Set Training or process training related to the use of the Tool Set, the results and estimates they gave might not be accurate. We recommend more usability testings after training was conducted to validate the results obtained in this study.

3.2 Capacity Planner Tool Set: General Usability

In response to Capacity Planner Tool Set Release 1.0, the majority of the responses were very positive on items that measured ease of use, usefulness, and the need for the Tool Set (refer to Figure 1).

In general, the participants found the format and presentation of the Tool Set easy to use. They also felt that Capacity Analyzer will help in obtaining and analyzing loop information. However, they were concerned about performance of the Tool Set and wanted performance improvement on planning request modification, retrieval and creation.

All the participants indicated that they strongly need an enhanced version of the Capacity Planner Tool Set that can recommend architecture, models and configurations as well as generating plans for planning tasks.

Table 1. Capacity Planner Tool Set General Usability and Need Index

Topics	Usability Score	Usability Grade
Capacity Analyzer is easy to use	3.27	B+
Capacity Analyzer helps in obtaining and analyzing loop information	3.45	B+
Topic	Need Score	Need Index
need enhanced version of Capacity Planner Tool Set	3.78	A (strong need)

Note: Conventions for usability grades or need index: A (above 3.75), A- (3.5 to 3.74), B+ (3.25 to 3.49), B (3.00 to 3.24) and B- (2.75 to 2.99). Usability scores range from 1 to 4 with 4 being the highest.

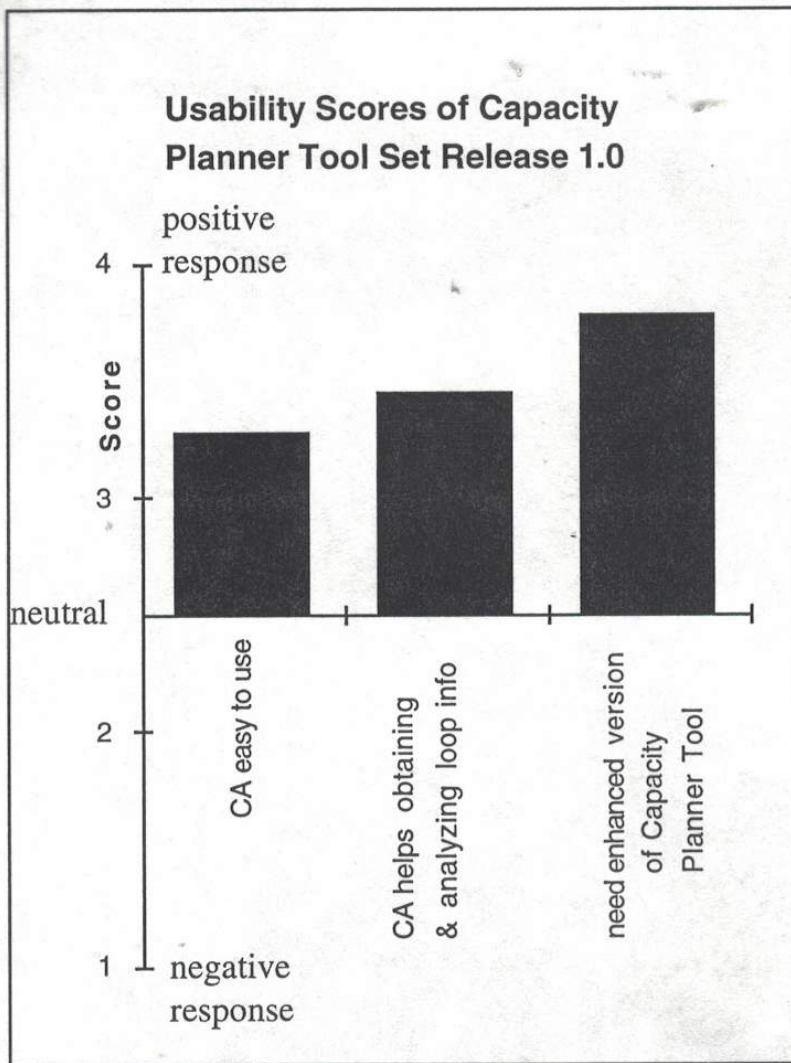


Figure 1. Capacity Planner Tool Set Release 1.0 general usability ratings.

3.3 Capacity Planner Tool Set : Planning Request Time Estimates

Most of the participants indicated that the use of Capacity Planner Tool Set will cut down their planning time substantially. Below are the survey data from the eleven participants (Table 2). Since there is a large variations of the estimated times, it is recommended that more specific testing and estimates to be done after training was completed and the users start to use the Tool Set in their work.

Table 2. Estimated time to perform planning request using current system vs. Capacity Planner Tool Set Release 1.0 (raw data)

Yrs of Experience in Tactical Planning	Job Type	Estimated time to obtain pertinent information using current systems and processes (hrs)	Estimated time to perform planning request using current systems and processes (hrs)	Estimated total time to perform planning request using current systems and processes (hrs)	Estimated time to perform planning request using Capacity Analyzer at this stage (hrs)
14	Tactical Planner	0.5	160	160.5	N/A
14	Prototype	8	40	48	8
15	Tactical Planner	0.5	16	16.5	40
1	Tactical Planner	40	40	80	20
9	Tactical Planner	4	16	20	8
22	Tactical Planner	12	12	24	3
0.5	Tactical Planner	80	40	120	20
-	Design Engineer	N/A	N/A	N/A	N/A
17	Tactical Planner	2	4	6	4
7	Tactical Planner	36	12	48	48
20	Prototype	N/A	N/A	N/A	1

A preliminary analysis was done on the data above excluding the three participants having "N/A" in their estimates (either they did not give the estimate or it did not apply). The results were shown in the following table and figure:

Table 3. Average Estimated Time between current systems and Capacity Planner Tool Set Release 1.0

Average estimated time to obtain pertinent information and time to perform planning request using current system and processes (hrs)	Average estimated time to perform planning request using Capacity Analyzer (CA) at this stage (hrs)	Estimated performance gain of productivity by using Capacity Analyzer
45.3	18.9	about 240%

A t-test showed that the differences were significant at $\alpha = 0.05$ ($t = 1.786$; $t_{critical} = 1.761$)

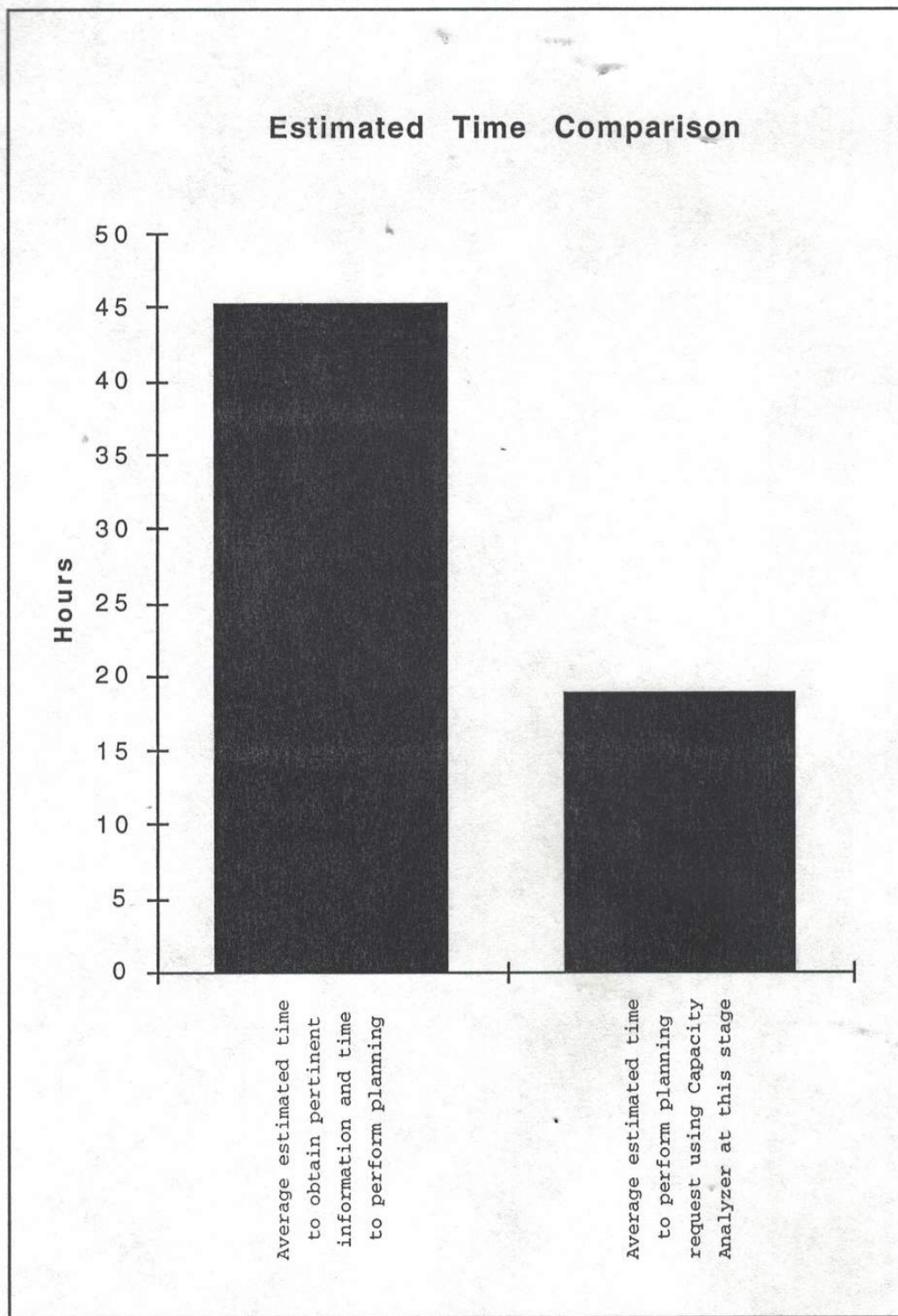


Figure 2. Comparison of estimated time required to perform planning request

3.4 Capacity Planner Tool Set User Interface : Individual Screen Ratings

In general, most of the screens of the Capacity Planner Tool Set were rated as very easy to use (refer to Table 4). We recommend redesigning some of the screens having ratings B or B- to further enhance the usability of the Tool Set.

Table 4. Capacity Planner Tool Set Release 1.0 : Screen ratings.

Screens	Usability Score	Usability Grade
asedit	3.17	B
Capacity Planner Main	3.32	B+
Filter Dialog Box - Capacity Planner Main	2.91	B
Multi-Sort Dialog Box - Capacity Planner Main	3.36	B+
Preference - Capacity Planner Main	2.92	B
New Planning Request Dialog Box - Capacity Planner Main	3.42	B+
Existing Planning Request Dialog Box - Capacity Planner Main	2.92	B
Select Wire Center Dialog Box - Capacity Planner Main	3.36	B+
Location Browser - Capacity Analyzer	3.37	B+
Loop Electronics Spreadsheet - Capacity Analyzer	3.56	A-
Add FP Dialog Box - Capacity Analyzer	3.62	A-
Adjust Threshold Dialog Box - Capacity Analyzer	3.52	A-
Multi-Sort Dialog Box - Capacity Analyzer	3.24	B
Filter Dialog Box - Capacity Analyzer	2.69	B-
Forecast Utilization Graph - Capacity Analyzer	2.92	B

3.5 Capacity Planner Tool Set - Comments and Suggestions

Below are some of the more general comments and suggestions taken from the usability questionnaires or informal interviews. Both positive and negative comments have been included.

Comments and Suggestions
very user friendly system.
system is useful only if database is current.
speed in retrieving or changing data need to be improved.
ability to generate reports will be good.
some of the terminology used need to be clarified.
more training will be needed
this is only 1/3 of the tool set

4.0 Conclusions

The overall feedback from the participants was favorable. Results from the usability scale revealed that the general attitude is positive but cautious. The results confirmed that Capacity Planner Tool Set Release 1.0 is in the right direction. The participants rated the majority of the windows as easy to use.

Although the estimated time should be interpreted with caution, preliminary results seemed to suggest that most of the tactical planners participating in the study estimated that there will be substantial time reduction in using Capacity Analyzer for planning. The average estimate showed a gain of about 240%. More study will be needed to validate these estimates after the Capacity Planner Tool Set is deployed and the tactical planners receive adequate training on its use.

The comments and suggestions from the users show that the tactical planners have a strong need of having an enhanced version of the Capacity Planner Tool Set which can recommend architecture, models and configurations based on the capacity shortages.

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